

Auteur: Noran Ben Said
Studierichting: Informatica
Oefeningenreeks 1C nr. 46.9

$$z^4 = 8 - 8i\sqrt{3}$$

$$r = \sqrt{8^2 + (-8\sqrt{3})^2} = 16$$

$$\theta = -\frac{\pi}{3}$$

$$8 - 8i\sqrt{3} = 16 \operatorname{cis} \left(-\frac{\pi}{3} \right)$$

$$z_k = \sqrt[4]{16} \operatorname{cis} \left(\frac{-\frac{\pi}{3} + k2\pi}{4} \right)$$

$$= 2 \operatorname{cis} \left(-\frac{\pi}{12} + k\frac{\pi}{2} \right)$$

$$z_0 = 2 \operatorname{cis} \left(-\frac{\pi}{12} \right)$$

$$z_1 = 2 \operatorname{cis} \frac{5\pi}{12}$$

$$z_2 = 2 \operatorname{cis} \frac{11\pi}{12}$$

$$z_3 = 2 \operatorname{cis} \frac{17\pi}{12}$$

References